

1. Core Entities			
Table: PATIENT			
Attribute Name	Data Type	Key	Description
patient_id	VARCHAR(36)	PK	Unique identifier for the patient (UUID).
patient_details_json	JSON		Stores demographics, contact info, and registration details.
Table: USER			
Attribute Name	Data Type	Key	Description
user_id	VARCHAR(36)	PK	Unique identifier for system users/staff.
username_string	VARCHAR(255)		Login username for the system.
user_type_enum	ENUM		Allowed values: 'ADMINISTRATOR', 'DOCTOR', 'NURSE', 'CASHIER', 'LAB_STAFF'.
2. Consent & Questionnaires			
Table: PATIENT_CONSENT			
Attribute Name	Data Type	Key	Description
consent_id	VARCHAR(36)	PK	Unique identifier for the consent form.
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).
authorization_type_str	VARCHAR(50)		Status of consent (e.g., 'active', 'inactive').
signature_binary	BLOB		Digital signature or biometric confirmation data.
timestamp_datetime	DATETIME		Date and time the consent was signed.
Table: PATIENT_QUESTIONNAIRE_RESPONSE			
Attribute Name	Data Type	Key	Description
response_id	VARCHAR(36)	PK	Unique identifier for the questionnaire submission.
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).
response_data_json	JSON		Structured pre-assessment data (e.g., reported symptoms).
completed_status_bool	BOOLEAN		True if completed, False if abandoned/partial.
timestamp_datetime	DATETIME		Date and time of submission.
3. Appointments & Queue Management			
Table: APPOINTMENT			
Attribute Name	Data Type	Key	Description
appointment_id	VARCHAR(36)	PK	Unique identifier for the appointment.
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).
requested_datetime	DATETIME		The scheduled date and time.
appointment_type_str	VARCHAR(50)		Allowed values: 'SCHEDULED', 'WALK_IN'.
assigned_doctor_user	VARCHAR(36)	FK	References USER(user_id). Must be a DOCTOR.
Table: ACTIVE_APPOINTMENT_QUEUE			
Attribute Name	Data Type	Key	Description
queue_entry_id	VARCHAR(36)	PK	Unique identifier for the queue log.
appointment_id	VARCHAR(36)	FK	References APPOINTMENT(appointment_id).
queue_number_string	VARCHAR(20)		The display number assigned to the patient (e.g., "A-012").
arrival_timestamp_datetime	DATETIME		The exact time the patient checked in.
4. Clinical & Diagnostics			
Table: CLINICAL_RECORD_ENCOUNTER			
Attribute Name	Data Type	Key	Description
clinical_record_id	VARCHAR(36)	PK	Unique identifier for the medical encounter.
appointment_id	VARCHAR(36)	FK	References APPOINTMENT(appointment_id).
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).
symptoms_text	TEXT		Free-form text of symptoms logged by staff.
vitals_json	JSON		Structured vital signs (BP, HR, Temp, Weight, etc.).
timestamp_datetime	DATETIME		Date and time the encounter started.
Table: ENCOUNTER_WORKFLOW_LOG			
Attribute Name	Data Type	Key	Description
workflow_entry_id	VARCHAR(36)	PK	Unique identifier for the workflow action.
clinical_record_id	VARCHAR(36)	FK	References CLINICAL_RECORD_ENCOUNTER(clinical_record_id).

performing_user_id	VARCHAR(36)	FK	References USER(user_id). Tracks who did the action.			
action_type_enum	ENUM		CHECK_IN', 'ASSESSMENT', 'PRESCRIBE', 'DIAGNOSE', 'CONSULT', 'BILLING'.			
timestamp_datetime	DATETIME		Exact time the action was performed.			
Table: DIAGNOSIS						
Attribute Name	Data Type	Key	Description			
diagnosis_id	VARCHAR(36)	PK	Unique identifier for the diagnosis.			
clinical_record_id	VARCHAR(36)	FK	References CLINICAL_RECORD_ENCOUNTER(clinical_record_id).			
diagnosing_doctor_us	VARCHAR(36)	FK	References USER(user_id).			
diagnosis_details_json	JSON		Detailed notes on the diagnosis.			
diagnosis_code_string	VARCHAR(50)		ICD-10 or SNOMED CT medical code.			
timestamp_datetime	DATETIME		Date and time of the diagnosis.			
Table: PRESCRIPTION						
Attribute Name	Data Type	Key	Description			
prescription_id	VARCHAR(36)	PK	Unique identifier for the prescription.			
clinical_record_id	VARCHAR(36)	FK	References CLINICAL_RECORD_ENCOUNTER(clinical_record_id).			
prescribing_doctor_us	VARCHAR(36)	FK	References USER(user_id).			
medication_details_json	JSON		Drug name, brand, strength, and quantity.			
dosage_details_string	VARCHAR(255)		Instructions (e.g., "Take 1 tablet every 8 hours").			
start_datetime	DATETIME		Start date of the medication.			
end_datetime	DATETIME		End date of the medication.			
timestamp_datetime	DATETIME		Date and time prescribed.			
Table: LABORATORY_REQUEST						
Attribute Name	Data Type	Key	Description			
lab_request_id	VARCHAR(36)	PK	Unique identifier for the lab order.			
clinical_record_id	VARCHAR(36)	FK	References CLINICAL_RECORD_ENCOUNTER(clinical_record_id).			
requesting_doctor_us	VARCHAR(36)	FK	References USER(user_id).			
request_test_code_str	VARCHAR(50)		LOINC or internal code for the test required.			
request_details_json	JSON		Specific instructions or panels requested.			
status_enum	ENUM		PENDING', 'COMPLETED'.			
timestamp_datetime	DATETIME		Date and time the request was made.			
Table: LABORATORY_RESULT						
Attribute Name	Data Type	Key	Description			
lab_result_id	VARCHAR(36)	PK	Unique identifier for the lab result.			
lab_request_id	VARCHAR(36)	FK	References LABORATORY_REQUEST(lab_request_id).			
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).			
encoded_by_user_id	VARCHAR(36)	FK	References USER(user_id) (Lab Staff).			
result_details_json	JSON		Structured lab data (biomarkers, values, reference ranges).			
timestamp_datetime	DATETIME		Date and time the results were encoded.			
5. AI Modules						
Table: AI_KNOWLEDGE_BASE_INSIGHT						
Attribute Name	Data Type	Key	Description			
ai_insight_id	VARCHAR(36)	PK	Unique identifier for the generalized insight.			
insight_pattern_details	JSON		The medical pattern or analytics data learned by the AI.			
validated_by_doctor_us	VARCHAR(36)	FK	References USER(user_id). Tracks which doctor approved the insight for the knowledge base.			
timestamp_datetime	DATETIME		Date and time generated.			
Table: PATIENT_ENCOUNTER_AI_LOG						
Attribute Name	Data Type	Key	Description			
ai_summary_log_id	VARCHAR(36)	PK	Unique identifier for the AI interaction log.			
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).			
clinical_record_id	VARCHAR(36)	FK	References CLINICAL_RECORD_ENCOUNTER(clinical_record_id).			
generating_ai_agent_id	VARCHAR(36)	FK	Identifier of the specific LLM/Agent version used.			
insight_summary_details	JSON		Structured AI inferences regarding the patient's data.			
medical_summary_text	TEXT		Human-readable AI summary output.			

timestamp_datetime	DATETIME		Date and time generated.				
6. Healthcare Information Exchange (HIE) & Corporate							
Table: HIE_PARTNER							
Attribute Name	Data Type	Key	Description				
hie_partner_id	VARCHAR(36)	PK	Unique identifier for the external partner/clinic.				
hie_partner_name_str	VARCHAR(255)		Name of the partner institution.				
hie_partner_exchange	ENUM		Protocol used: 'FHIR', 'HL7', 'CDA'.				
Table: HIE_TRANS_LOG							
Attribute Name	Data Type	Key	Description				
hie_transaction_id	VARCHAR(36)	PK	Unique identifier for the data exchange event.				
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).				
hie_partner_id	VARCHAR(36)	FK	References HIE_PARTNER(hie_partner_id).				
shared_medical_record_id	VARCHAR(255)		Endpoint URL or reference to the transmitted data.				
authorizing_consent_id	VARCHAR(36)	FK	References PATIENT_CONSENT(consent_id).				
sharing_datetime	DATETIME		Exact timestamp of the data transfer.				
Table: CORPORATE_CLIENT							
Attribute Name	Data Type	Key	Description				
corporate_client_id	VARCHAR(36)	PK	Unique identifier for the corporate client (e.g., an insurance company or employer).				
corporate_client_name	VARCHAR(255)		Name of the corporation.				
Table: CORPORATE_REQUEST_LOG							
Attribute Name	Data Type	Key	Description				
corporate_request_id	VARCHAR(36)	PK	Unique identifier for the corporate data request.				
corporate_client_id	VARCHAR(36)	FK	References CORPORATE_CLIENT(corporate_client_id).				
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).				
hie_request_details_json	JSON		Details of what information the corporation is requesting.				
request_status_enum	ENUM		PENDING', 'APPROVED', 'DENIED'.				
authorizing_consent_id	VARCHAR(36)	FK	References PATIENT_CONSENT(consent_id).				
timestamp_datetime	DATETIME		Date and time the request was lodged.				
7. Billing & Payment							
Table: BILLING_RECORD							
Attribute Name	Data Type	Key	Description				
invoice_id	VARCHAR(36)	PK	Unique identifier for the generated invoice.				
patient_id	VARCHAR(36)	FK	References PATIENT(patient_id).				
clinical_record_id	VARCHAR(36)	FK	References CLINICAL_RECORD_ENCOUNTER(clinical_record_id).				
total_bill_amount_decimal	DECIMAL(10,2)		Total amount due for the encounter.				
timestamp_datetime	DATETIME		Date and time the invoice was generated.				
Table: PAYMENT_RECORD							
Attribute Name	Data Type	Key	Description				
payment_id	VARCHAR(36)	PK	Unique identifier for the payment transaction.				
invoice_id	VARCHAR(36)	FK	References BILLING_RECORD(invoice_id).				
cashier_user_id	VARCHAR(36)	FK	References USER(user_id). Tracks who processed it.				
payment_details_json	JSON		Payment method (cash, card, insurance), reference numbers.				
timestamp_datetime	DATETIME		Date and time the payment was completed.				